

VERIHIRE

AI-POWERED SKILL CERTIFICATION AND HIRING PLATFORM

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INTRODUCTION

- Traditional resumes and self-claimed skills lack credibility.
- Recruiters face hiring risks due to resume fraud and unverifiable claims.
- Skill certifications today are often theoretical and do not reflect real-world ability.
- There is a growing need for a transparent, merit-based system where skills are validated practically.

PROBLEM STATEMENT

Verihire combats resume fraud by using AI-generated skill challenges with automated scoring and peer validation, creating verified portfolios for candidates and giving recruiters a transparent, skill-based hiring dashboard.

OBJECTIVES

- Verify Real-World Skills – Use AI-generated challenges to test practical abilities beyond resumes.
- Automate Skill Evaluation – Apply AI models for unbiased, consistent scoring.
- Build Dynamic Portfolios – Provide candidates with auto-updated, skill-based profiles.
- Streamline Hiring – Offer recruiters a dashboard to filter and shortlist based on verified skills.

KEY CHALLENGES

- Evaluating AI models on real-world vocational skills is limited.
- Lack of trust and transparency in AI certification processes.
- Peer grading suffers from bias, collusion, and low-effort reviews.
- Digital certificates are vulnerable to forgery and verification issues.

EXISTING SOLUTIONS

- Benchmark datasets (e.g., Professional Certification Benchmark) for AI exam-style testing.
- Governance frameworks (e.g., AI Certification by Cihon et al.) to reduce info asymmetry.
- Blockchain-based platforms (e.g., SkillCheck) for securing the certificates.
- Peer-grading systems (e.g., CrowdGrader) with reputation-weighted scoring.
- Blockchain credential standards (e.g., Blockcerts) for tamper-proof certificates

LITERATURE REVIEW

Reference	Methodology	Drawbacks
D. Noever and M. Ciolino, "Professional Certification Benchmark Dataset: The First 500 Jobs for Large Language Models," PeopleTec, Inc., Huntsville, AL, USA, Tech. Rep., May 2023.	Built benchmark dataset of certification/exam items; evaluated LLMs (GPT-3/3.5) on vocational readiness using quantitative scoring	Focuses on LLM QA performance; doesn't reflect real-world hands-on skill certification
BP. Cihon, M. Kleinaltenkamp, J. Schuett, and S. D. Baum, "AI Certification: Advancing Ethical Practice by Reducing Information Asymmetries," Global Catastrophic Risk Institute (GCRI), Working Paper 21-1, 2021	Conceptual / policy analysis; proposes AI certification principles and governance pathways	High-level guidance; lacks concrete technical implementation or automated scoring

J. Gupta and S. Nath, “SkillCheck: An Incentive-based Certification System using Blockchains,” Indian Institute of Technology Kanpur, Technical Report, 2020.	Certificates are issued digitally and anchored on blockchain for secure, tamper-proof verification. Learners and evaluators are incentivized with rewards to motivate participation.	Requires proper blockchain infrastructure for deployment. Assessment quality depends on platform design; blockchain ensures security but not validity
MIT Media Lab and Learning Machine, “Blockcerts: An Open Infrastructure for Academic Credentials on the Blockchain,” Whitepaper, 2016.	Blockchain-based verifiable credential standard; defines issuing, storing, verifying workflows	Secures certificates but not assessment quality; privacy, interoperability, and cost issues
L. de Alfaro and M. Shavlovsky, “CrowdGrader: Crowdsourcing the Evaluation of Homework Assignments,” in Proceedings of the 2013 ACM Conference on Learning at Scale, San Jose, CA, USA, pp. 1–10.2013	Peer-grading system for assignments; iterative reputation-weighted aggregation; field experiments	Reviewer bias, collusion, or low-effort reviews may reduce reliability; depends on good incentives

VERIHIRE Project Timeline

(Estimated Duration)



CONCLUSION

Verihire offers a transparent, AI-powered platform that validates real-world skills through automated evaluation and peer review.

It reduces hiring risks, combats resume fraud, and empowers both candidates and recruiters in building a merit-based ecosystem

REFERENCES

- [1] D. Noever and M. Ciolino, "Professional Certification Benchmark Dataset: The First 500 Jobs for Large Language Models," PeopleTec, Inc., Huntsville, AL, USA, Tech. Rep., May 2023.
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- [3] J. Gupta and S. Nath, "SkillCheck: An Incentive-based Certification System using Blockchains," Indian Institute of Technology Kanpur, Technical Report, 2020.



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[5] L. de Alfaro and M. Shavlovsky, “CrowdGrader: Crowdsourcing the Evaluation of Homework Assignments,” in Proceedings of the 2013 ACM Conference on Learning at Scale, San Jose, CA, USA, pp. 1–10. 2013

Thank You